

LITTLE FALLS, MN, USA

WMO: 726578

Lat: 45.949N

Lon: 94.347W

Elev: 342

StdP: 97.28

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04944

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.7	-24.8	-32.3	0.2	-27.4	-28.9	0.3	-23.3	8.8	-7.7	8.0	-9.2	0.9	290	0.556

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	32.1	22.6	29.8	21.0	27.7	19.7	24.0	28.8	22.8	27.6	21.7	26.1	4.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.5	17.9	26.7	21.3	16.6	25.6	20.1	15.4	24.4	74.1	28.9	69.1	27.8	64.6	26.1	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.4	7.5	6.6	-32.2	34.3	3.6	2.0	-34.7	35.8	-36.8	36.9	-38.8	38.0	-41.4	39.4
			-32.4	25.9	3.2	1.7	-34.7	27.1	-36.6	28.1	-38.4	29.1	-40.8	30.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.9	-10.9	-8.7	-1.5	5.9	12.8	18.3	20.9	19.3	15.0	7.6	-0.4	-7.9
	DBStd	12.38	7.46	7.35	6.72	5.40	4.66	3.85	3.28	3.23	4.75	4.97	5.90	6.88
	HDD10.0	2689	647	524	361	143	23	1	0	0	10	108	316	556
	HDD18.3	4753	906	758	616	374	181	46	12	27	122	334	563	814
	CDD10.0	1203	0	0	3	20	111	250	337	287	160	33	2	0
	CDD18.3	224	0	0	0	1	9	45	91	56	21	1	0	0
	CDH23.3	2362	0	0	0	13	151	479	930	560	211	18	0	0
	CDH26.7	666	0	0	0	2	41	133	294	145	49	2	0	0
Wind	WSAvg	2.5	2.5	2.5	2.7	2.9	2.9	2.5	2.2	2.1	2.4	2.8	2.7	2.4
Precipitation	PrecAvg	703	18	18	40	65	77	112	97	82	78	69	33	20
	PrecMax	900	60	42	102	178	159	183	221	182	156	166	110	45
	PrecMin	493	0	3	15	17	23	55	27	18	22	22	0	5
	PrecStd	97	15	12	23	38	35	38	42	42	36	38	29	13
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.4	9.9	18.9	25.2	31.0	33.1	34.0	32.7	31.2	26.0	17.7	7.4
		MCWB	2.9	6.0	12.7	13.7	19.0	23.0	24.2	23.7	21.0	16.9	10.8	4.1
	2%	DB	2.5	6.1	14.1	21.3	27.3	30.0	32.2	30.1	27.7	22.1	13.7	4.1
		MCWB	0.6	3.4	9.1	11.7	16.5	20.8	23.4	21.4	19.6	15.2	8.9	2.0
	5%	DB	0.9	3.0	10.9	17.9	24.3	27.7	30.0	27.8	26.1	18.0	10.8	2.3
		MCWB	-0.1	1.4	7.6	9.8	14.9	19.0	21.8	20.1	18.3	12.7	6.9	1.0
	10%	DB	-0.9	1.4	7.5	16.0	22.2	26.2	27.7	27.0	23.0	16.1	7.7	0.9
		MCWB	-1.5	0.2	4.6	8.7	14.4	18.2	20.3	19.6	17.3	11.3	5.0	0.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.3	6.6	13.9	15.8	21.3	24.7	26.4	25.3	23.0	19.2	13.0	5.0
		MCDB	6.3	8.7	17.4	22.8	28.3	30.6	30.7	30.7	27.3	22.9	15.9	7.3
	2%	WB	0.8	3.8	10.1	12.8	18.6	22.7	24.6	23.2	21.5	16.1	9.7	2.6
		MCDB	1.9	5.7	13.2	18.9	24.2	27.9	29.6	27.7	25.9	19.9	13.2	4.2
	5%	WB	-0.3	1.9	7.3	11.0	16.9	21.2	23.2	21.9	20.1	13.8	7.4	1.0
		MCDB	0.5	3.4	10.0	17.2	21.7	25.9	28.1	26.2	24.0	17.9	10.1	1.9
	10%	WB	-1.5	0.3	4.5	9.1	15.3	19.7	22.0	20.7	18.7	11.8	5.2	-0.1
		MCDB	-0.8	1.3	7.4	14.9	20.5	24.0	26.5	24.3	22.3	14.9	8.0	0.5
Mean Daily Temperature Range	5% DB	MDBR	10.0	10.7	10.1	11.7	12.5	11.8	12.3	12.4	12.1	10.2	8.8	8.5
		MCDBR	10.0	10.6	14.0	16.9	16.8	14.2	14.0	14.2	14.8	14.6	13.6	10.2
		MCWBR	8.3	8.4	9.0	9.2	8.0	6.6	6.6	6.9	7.5	8.4	8.9	8.2
	5% WB	MCDBR	9.3	9.8	12.9	14.6	13.2	11.6	12.4	11.9	11.8	12.4	12.0	8.9
		MCWBR	8.0	7.8	8.8	8.9	7.5	6.4	6.7	6.8	7.0	8.1	8.4	7.5
Clear-Sky Solar Irradiance	taub	0.261	0.282	0.310	0.365	0.394	0.386	0.382	0.394	0.361	0.312	0.283	0.259	
	taud	2.388	2.324	2.375	2.305	2.281	2.343	2.377	2.347	2.415	2.539	2.506	2.423	
	Ebn at Noon	857	899	921	892	873	880	879	854	855	861	827	820	
	Edh at Noon	71	93	103	121	129	122	116	115	99	75	64	62	
All-Sky Solar Radiation	RadAvg	1.67	2.77	3.82	4.75	5.38	5.94	6.33	5.29	4.01	2.51	1.58	1.16	
	RadStd	0.20	0.30	0.36	0.41	0.41	0.40	0.24	0.44	0.30	0.37	0.15	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	-1.12	-1.12	-1.42	N/A	N/A	N/A	-72
(47)	Regional (60 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page